

Lightweight CNN Architecture for Bangla Handwritten Digit Recognition

August 8, 2026

1 Model Architecture

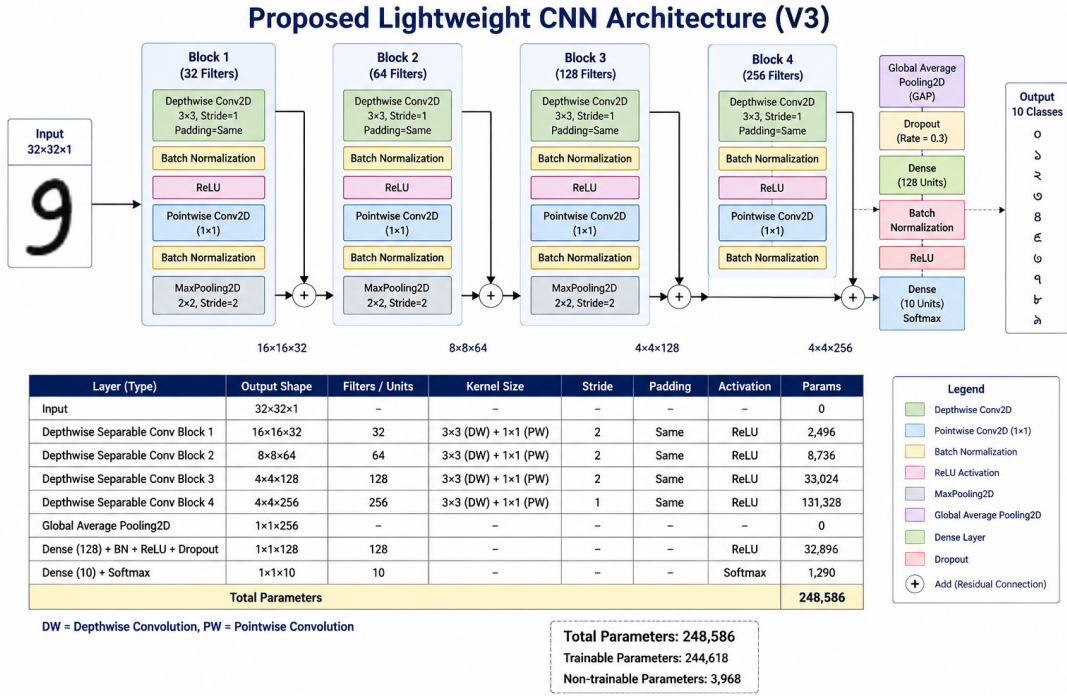


Figure 1: Proposed Lightweight CNN Architecture

1.1 Dataset Split Ratio

After preprocessing and removing 18 exact duplicate images, the final NumtaDB dataset contained 72,027 unique handwritten digit images. To ensure that the class distribution was preserved across all subsets, the dataset was divided using **stratified sampling**. The dataset was split into **70% training**, **15% validation**, and **15% testing**.

Table 1: Dataset Split Ratio

Dataset	Number of Images	Ratio
Training	50,418	70%
Validation	10,804	15%
Testing	10,805	15%
Total	72,027	100%

1.2 Training History

Table 2: Training and Validation Performance of the Proposed Lightweight CNN

Epoch	Train Acc.	Train Loss	Val. Acc.	Val. Loss	Learning Rate
1	79.36%	0.6219	64.51%	1.4797	0.001000
2	94.53%	0.1749	68.17%	1.2930	0.001000
3	95.72%	0.1376	53.74%	1.8750	0.001000
4	96.28%	0.1208	85.45%	0.4808	0.001000
5	96.73%	0.1050	48.39%	3.4994	0.001000
6	96.94%	0.0993	72.44%	1.1810	0.001000
7	97.24%	0.0877	76.60%	0.8584	0.001000
8	97.98%	0.0688	84.32%	0.5081	0.000500
9	98.03%	0.0655	92.35%	0.2237	0.000500
10	98.12%	0.0602	92.69%	0.2363	0.000500
11	98.18%	0.0594	66.61%	1.7269	0.000500
12	98.19%	0.0574	78.78%	0.7125	0.000500
13	98.28%	0.0553	92.10%	0.2443	0.000500
14	98.63%	0.0450	73.75%	1.1692	0.000250
15	98.70%	0.0402	78.77%	0.8695	0.000250
16	98.75%	0.0410	94.86%	0.1601	0.000250
17	98.72%	0.0416	96.21%	0.1283	0.000250
18	98.79%	0.0400	66.54%	2.0701	0.000250
19	98.85%	0.0380	98.39%	0.0525	0.000250
20	98.84%	0.0371	70.04%	1.3214	0.000250
21	98.86%	0.0360	62.80%	2.2693	0.000250
22	98.84%	0.0367	76.19%	1.0992	0.000250
23	98.97%	0.0312	97.44%	0.0796	0.000125
24	99.05%	0.0295	98.21%	0.0597	0.000125
25	99.07%	0.0290	96.61%	0.1152	0.000125
26	99.16%	0.0266	97.18%	0.0972	0.0000625
27	99.17%	0.0261	97.85%	0.0687	0.0000625

1.3 Test Set Evaluation

Table 3.

Table 3: Test Set Performance of the Proposed Lightweight CNN

Evaluation Metric	Result
Test Loss	0.0525
Test Accuracy	98.5007%

2 Training and Validation Performance

The model was trained using the NumtaDB dataset. Figures 2 and 3 present the training and validation accuracy and loss curves, respectively.

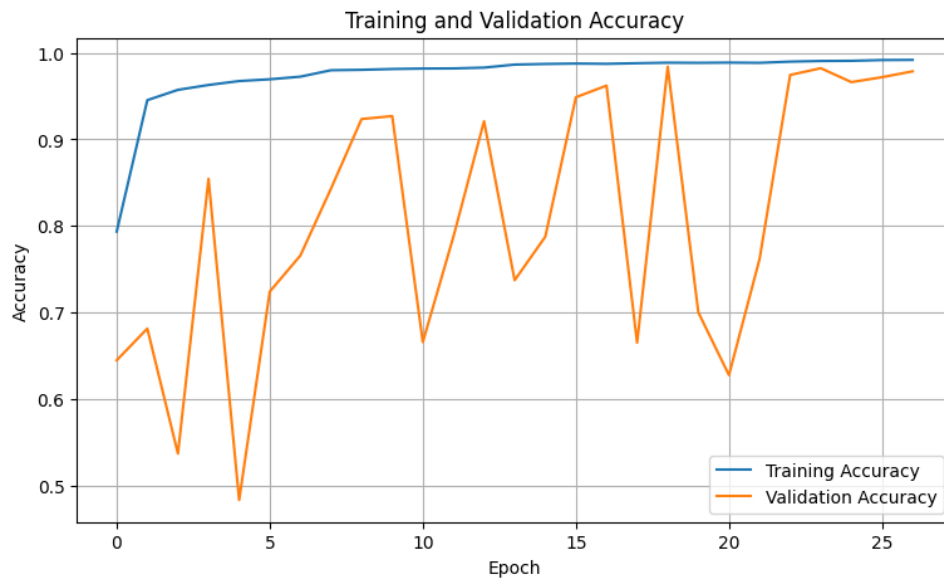


Figure 2: Training and Validation Accuracy

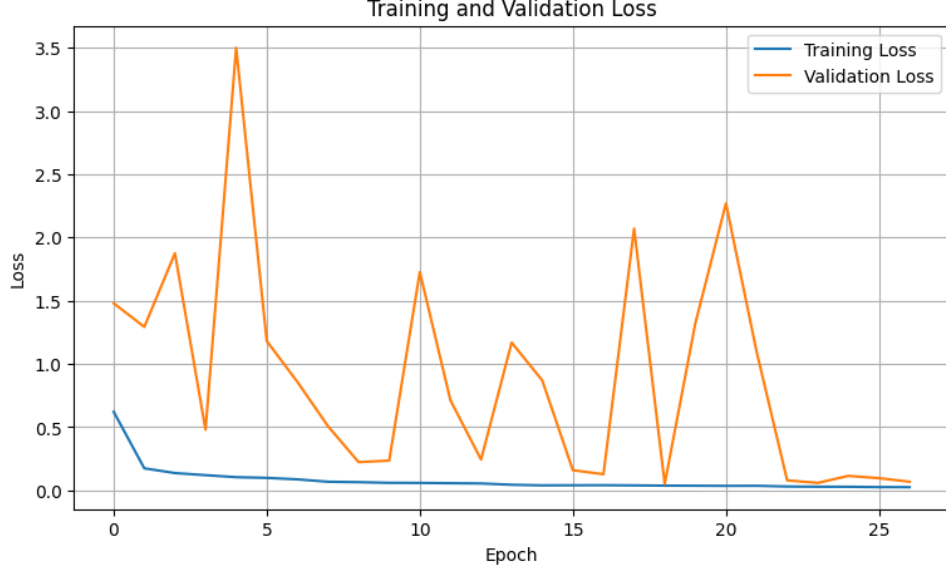


Figure 3: Training and Validation Loss

2.1 Preprocessing and Data Augmentation

The following preprocessing techniques were applied before training:

- Resize all images to 32×32 pixels.
- Convert RGB images to grayscale.
- Normalize pixel values to the range $[0, 1]$.
- Remove exact duplicate samples from the dataset.

The following data augmentation techniques were used:

- Random rotation.
- Random width shift.
- Random height shift.
- Random zoom.

2.2 Training Hyperparameters

The proposed lightweight CNN model was trained using the hyperparameters listed in Table 4.

Table 4: Training Hyperparameters

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	64
Maximum Epochs	50
Loss Function	Sparse Categorical Crossentropy
Activation Function	ReLU, Softmax
Input Size	$32 \times 32 \times 1$
Callbacks	EarlyStopping, ReduceLROnPlateau, ModelCheckpoint
EarlyStopping Patience	8
ReduceLROnPlateau Patience	3
Dropout Rate	0.30

3 Confusion Matrix

The confusion matrix shown in Figure 4 demonstrates that the proposed model correctly classifies most Bangla handwritten digits, indicating balanced performance across all classes.

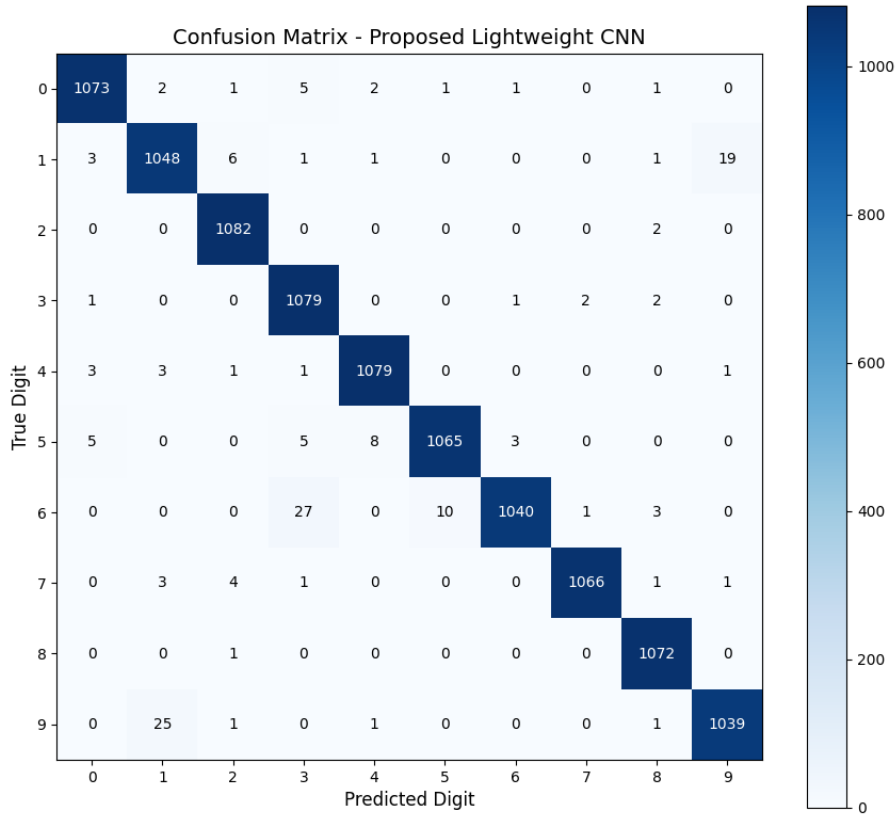


Figure 4: Confusion Matrix of the Proposed Model

3.1 Classification Report

The classification performance of the proposed lightweight CNN was evaluated using precision, recall, F1-score, and support for each Bangla handwritten digit class. The detailed classification results are presented in Table 5.

Table 5: Classification Report on the Test Set

Digit	Precision	Recall	F1-Score	Support
0	0.9889	0.9880	0.9885	1086
1	0.9695	0.9713	0.9704	1079
2	0.9872	0.9982	0.9927	1084
3	0.9643	0.9945	0.9791	1085
4	0.9890	0.9917	0.9904	1088
5	0.9898	0.9807	0.9852	1086
6	0.9952	0.9621	0.9784	1081
7	0.9972	0.9907	0.9939	1076
8	0.9898	0.9991	0.9944	1073
9	0.9802	0.9738	0.9770	1067
Accuracy			0.9850	10805
Macro Avg.	0.9851	0.9850	0.9850	10805
Weighted Avg.	0.9851	0.9850	0.9850	10805

4 Real Image Prediction

[b]0.45

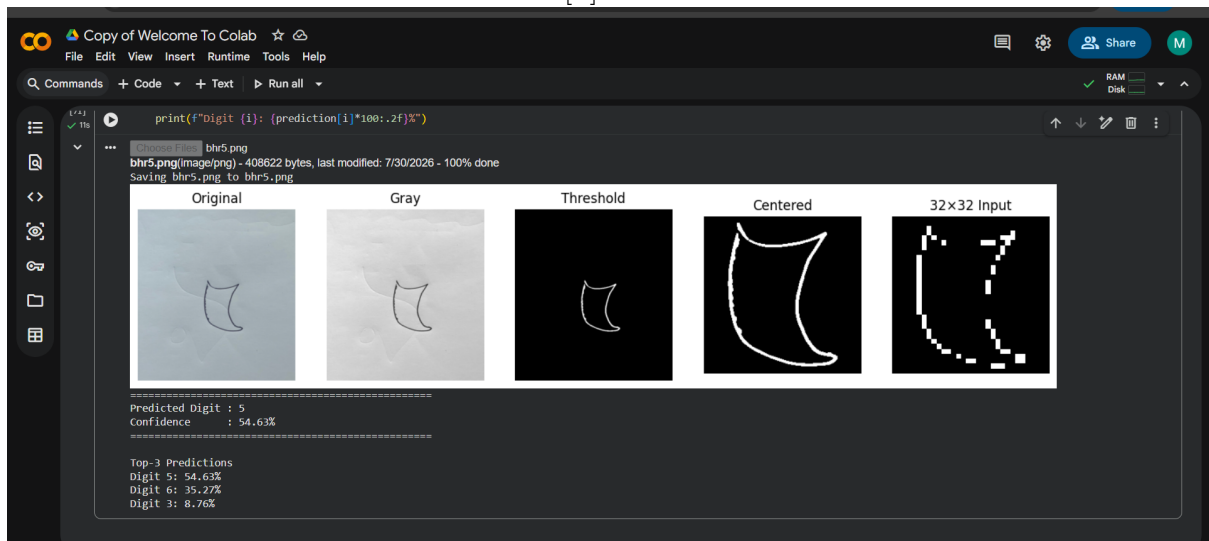


Figure 5: Original handwritten image

[b]0.45

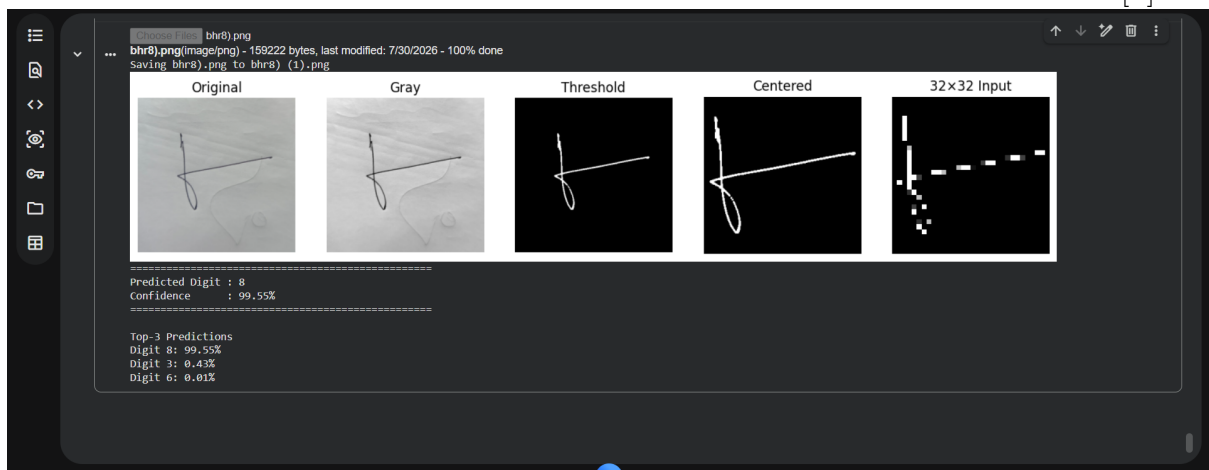


Figure 6: Prediction result